
بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

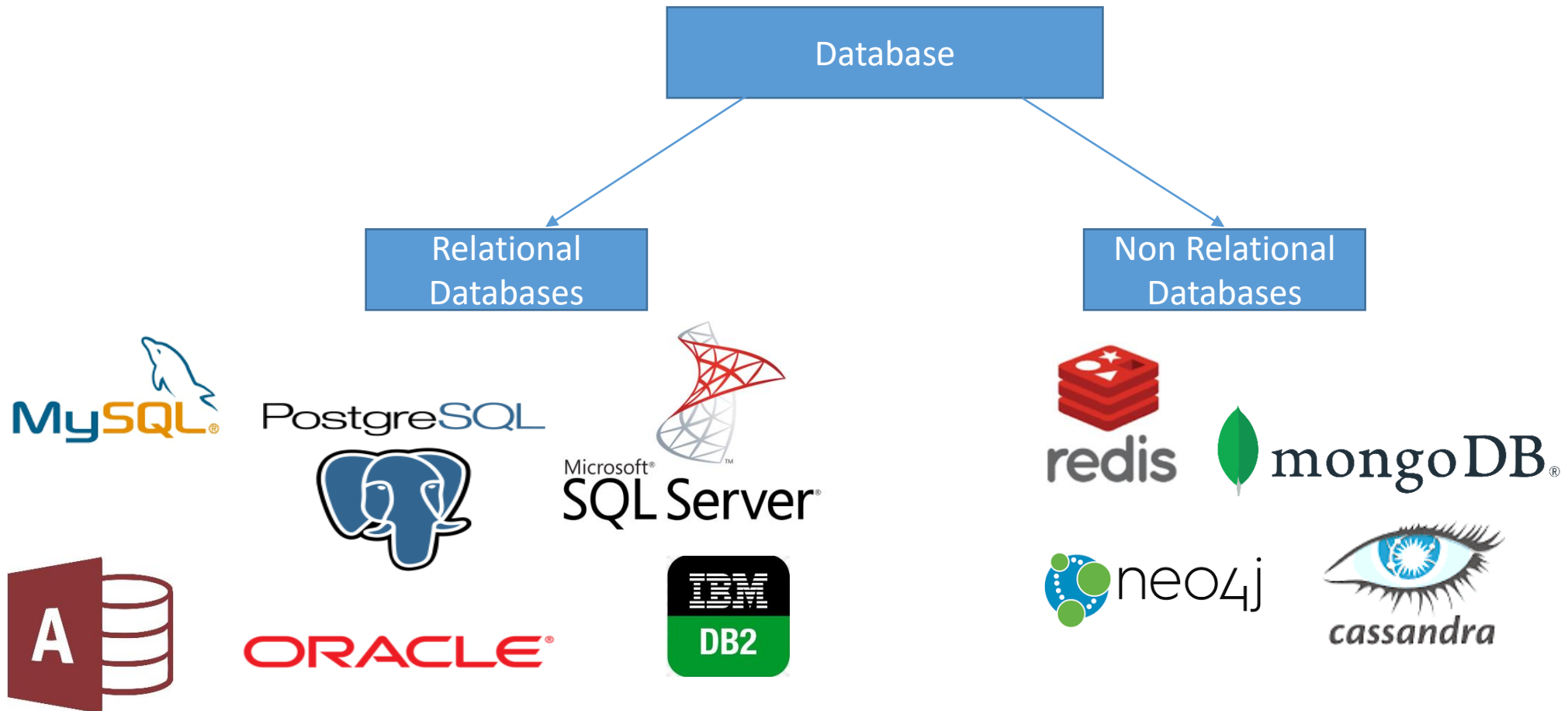
رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ
عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي

SQL Commands - DDL

Lecture # 3

Course Instructor: Waqar Ashiq

Types of Databases

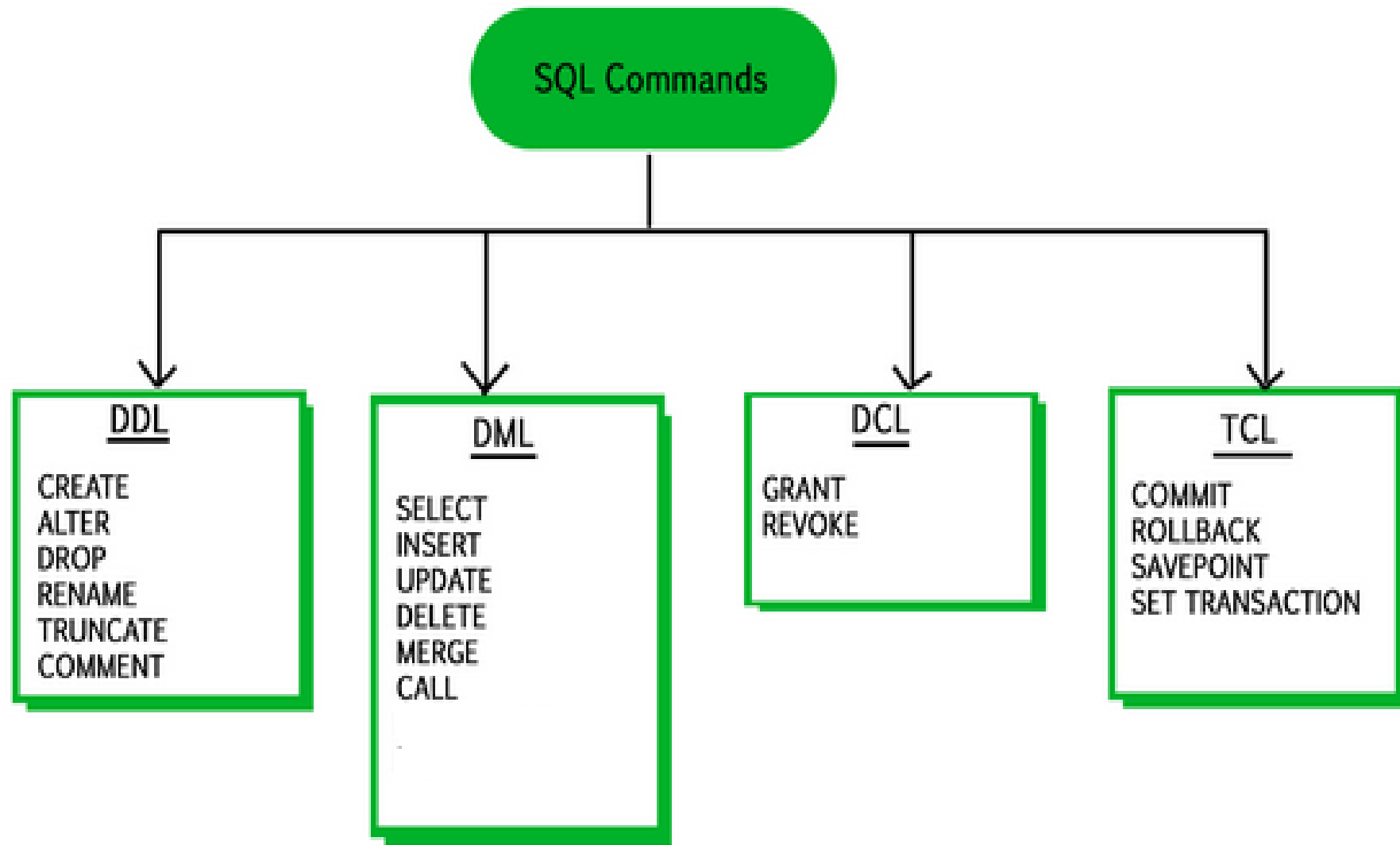


MySQL

- Open Source relational database management system
- Uses the SQL (Structured Query Language)
- A leading database for web applications
- Used for small apps to large enterprise apps
- Used with multiple languages (Java, PHP, Node, Python, C# etc.)
- Cross platform

Management Tools

- Desktop tools: MySQL Workbench
- Web based tool: PHPmyAdmin



DDL and DML

- These SQL commands are mainly categorized into four categories as discussed below:
- **DDL(Data Definition Language)** : DDL or Data Definition Language actually consists of the SQL commands that can be used to define the database schema. It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in database. **Examples of DDL commands:**
 - **CREATE** – is used to create the database or its objects
 - **DROP** – is used to delete objects from the database.
 - **ALTER**-is used to alter the structure of the database.
 - **TRUNCATE**–is used to remove all records from a table, including all spaces allocated for the records are removed.
 - **COMMENT** –is used to add comments to the data dictionary.
 - **RENAME** –is used to rename an object existing in the database.
- **DML(Data Manipulation Language)** : The SQL commands that deals with the manipulation of data present in database belong to DML or Data Manipulation Language and this includes most of the SQL statements. **Examples of DML:**
 - **SELECT** – is used to retrieve data from the a database.
 - **INSERT** – is used to insert data into a table.
 - **UPDATE** – is used to update existing data within a table.
 - **DELETE** – is used to delete records from a database table.

DCL and TCL

- **DCL(Data Control Language)** : DCL includes commands such as GRANT and REVOKE which mainly deals with the rights, permissions and other controls of the database system. **Examples of DCL commands:**
 - **GRANT**-gives user's access privileges to database.
 - **REVOKE**-withdraw user's access privileges given by using the GRANT command.
- **TCL(transaction Control Language)** : TCL commands deals with the transaction within the database. **Examples of TCL commands:**
 - **COMMIT**– commits a Transaction.
 - **ROLLBACK**– rollbacks a transaction in case of any error occurs.
 - **SAVEPOINT**–sets a save point within a transaction.

- **Relational Database Management System (RDBMS)**
- RDBMS is a software that store data in the form of relations(tables)
- **What is Table ?**
- A table is a data structure that store data in the form of rows and cols
- Cols are fields and row are records
- table is basic fundamental unit of storage in DB

- **Example:**

- A university Database

- Table : **Student**

- **Fields:**

- **Type:**

- **Type:**

- **Student**

Sid	Sname	Address	Gender	Marks	email
int	string	string	char	int	string
int	varchar(20)	varchar(40)	varchar(6)	float	varchar(30)

- Sid sname Address Gender Marks email

- 1 Farooq Lhr M 70 abc@h.com

- 2 Ahmed ISB M 85 xyz@h.com

- 3 ZAhid Lhr M 40 uvw@h.com

- **Primary Key Col**

- - Every rows should has unique identity

- - Every table should has primary key col

MY SQL

- **RDMBS** developed by ORACLE Corporation

- **Table Operations**

- 1- Insert
 - 2- Delete
 - 3- Update
 - 4- Query

- **What is Query?**

- Query is a question about data from database.
 - All databases used a special language for Query i.e. SQL (Structured Query Language)

SQL-DDL

- Structured Query Language.
- SQL is a standard language for accessing and manipulating databases.

- **Create Database Statement**

The CREATE DATABASE statement is used to create a new SQL database.

SQL Syntax:

Create database *databasename*

Create database hotel;

Create database gym;

SQL-DDL (cont...)

- **Create table statement**
- The create table statement is used to create a new table in a database.

Syntax:

Create Table Create Table

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,  
    column3 datatype,  
    ....  
);
```

Create Table - SQL

```
CREATE TABLE Member(  
  MemID int Primary Key,  
  LastName varchar(10),  
  FirstName varchar(15),  
  Address varchar(25),  
  City varchar(20)  
);
```

MemID	LastName	FirstName	Address	City

Exercise # 1

- Create a table called *supplier* which has 3 columns.
- The first column is called *supplier_id*.
- The second column is called *supplier_name*.
- The third column is called *contact_name*

Solution

```
CREATE TABLE supplier(  
  supplier_id int Primary Key,  
  supplier_name varchar(20),  
  contact_name varchar(20)  
);
```


Exercise# 2

- Create a SQL table called *customer* that stores customer ID, name, address, city, state and zip code information.

Solution Exercise 2

```
CREATE TABLE customer(  
customer_id int Primary Key,  
customer_name varchar(50),  
address varchar(50),  
city varchar(50),  
state varchar(50),  
zip_code varchar(50)  
);
```

SQL-DDL (cont...)

- **Drop database statement**

- The DROP DATABASE statement is used to drop an existing SQL database.

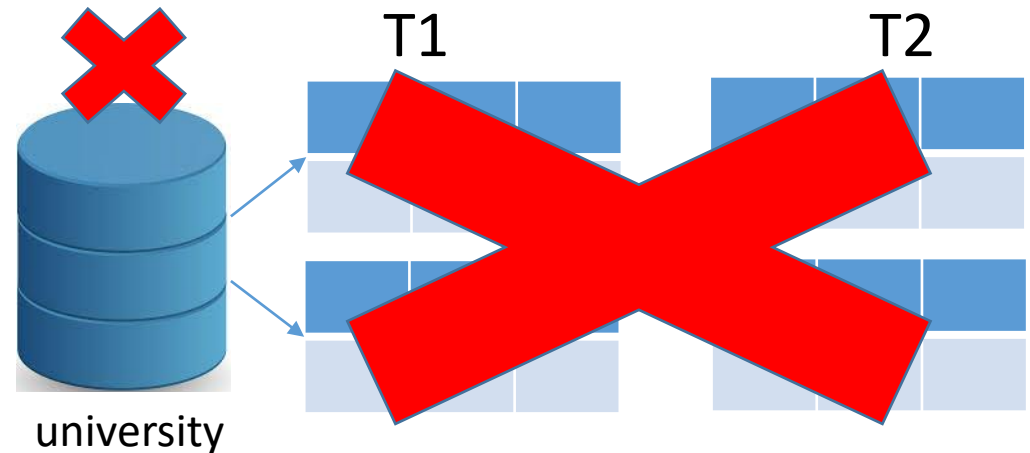
- **Syntax:**

DROP DATABASE *database_name*;

- **Example:**

Drop database university;

- **Note:** Be careful before dropping a database. Deleting a database will result in loss of complete information stored in the database!



SQL-DDL (cont...)

- **Drop table statement**

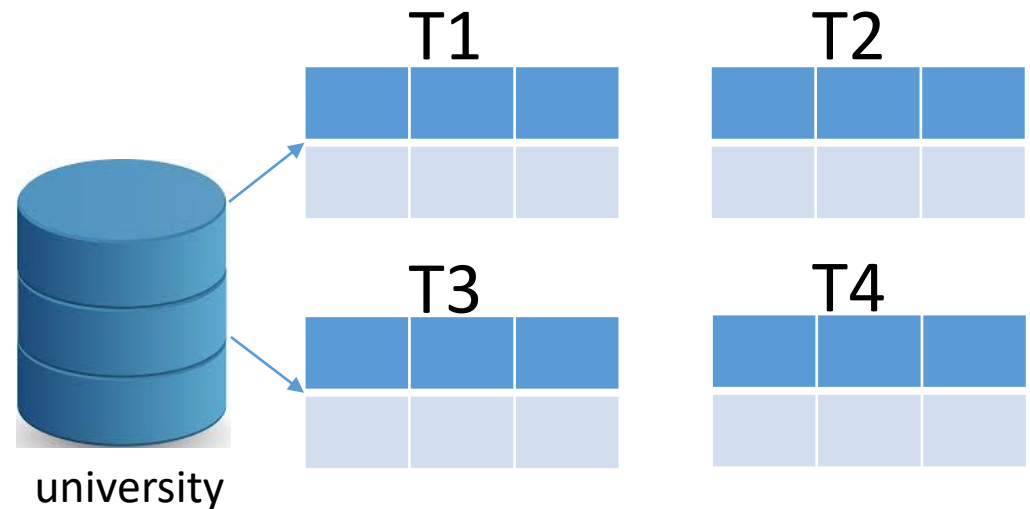
- The DROP table statement is used to drop an existing table in database.

- **Syntax:**

DROP table *table_name*;

- **Example:**

Drop table T1;



SQL-DDL (cont...)

- **Drop table statement**

- The DROP table statement is used to drop an existing table in database.

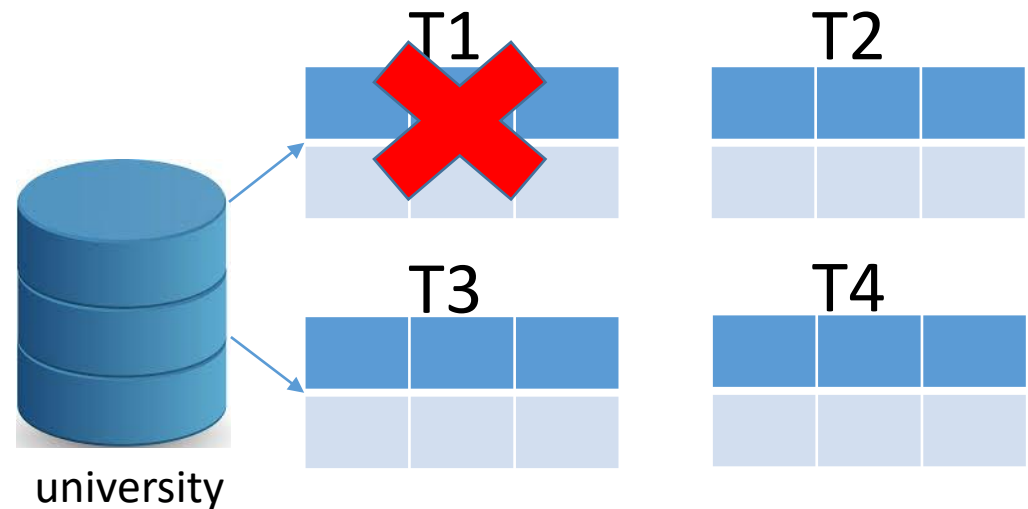
- **Syntax:**

DROP table *table_name*;

- **Example:**

Drop table T1;

- **Note:** Be careful before dropping a table. Deleting a table will result in loss of complete information stored in the table!



SQL-DDL (cont...)

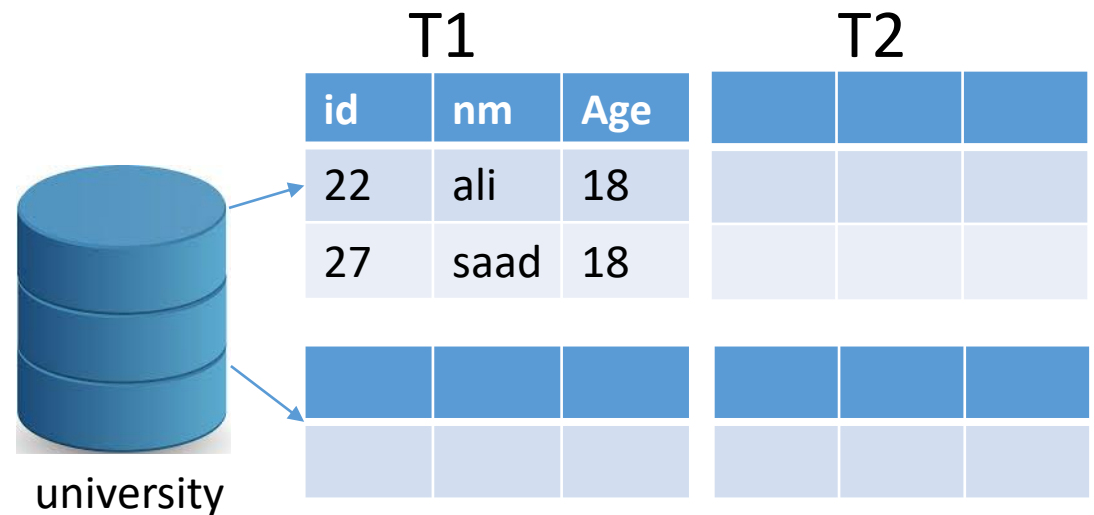
- **SQL Truncate table**
- The truncate table statement is used to delete the data inside a table, but not the table itself.

- **Syntax:**

TRUNCATE *table_name*;

- For example,

Truncate T1;



SQL-DDL (cont...)

- **SQL Truncate table**

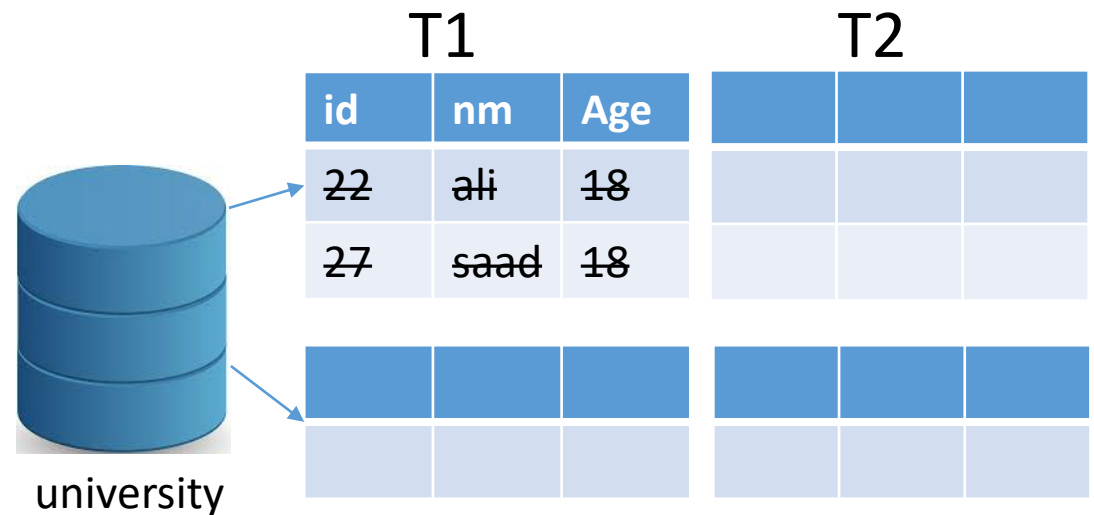
- The truncate table statement is used to delete the data inside a table, but not the table itself.

- **Syntax:**

TRUNCATE *table_name*;

- For example,

Truncate T1;



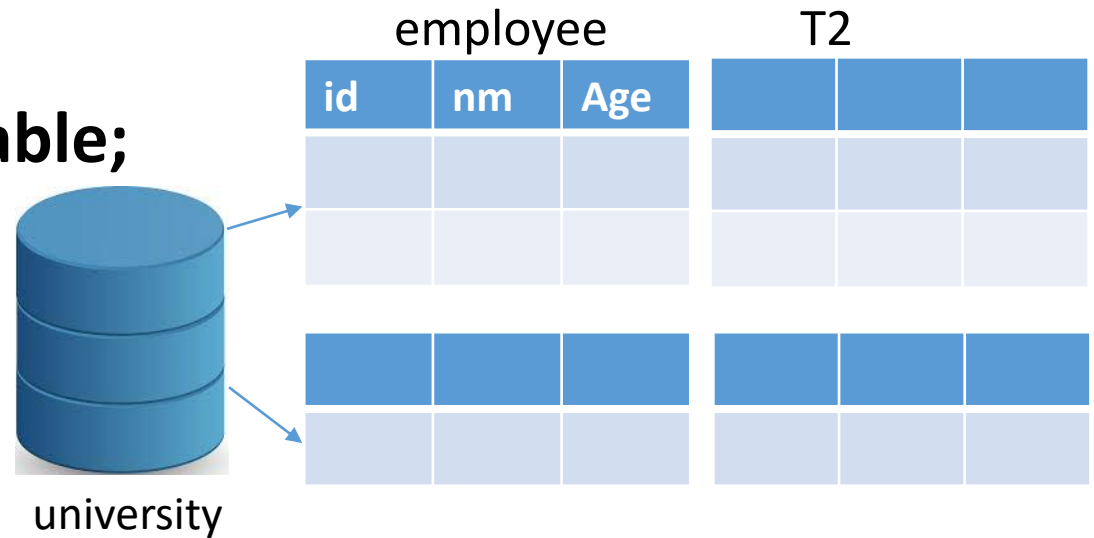
SQL-DDL (cont...)

- **Rename SQL statement**
- RENAME table renames one or more tables.
- **Syntax:**

RENAME TABLE old_table TO new_table;

- For example

Rename table T1 to employee;



SQL ALTER TABLE Statement

- The ALTER TABLE statement is used to add, delete, or modify columns in an existing table.
- **ALTER TABLE - ADD a Column**
- To add a column in a table, use the following syntax:
- **Syntax**

ALTER TABLE *table_name* ADD *column_name* *datatype*;

SQL ALTER TABLE Statement

Emp_ID	Emp_Name	City	Phone_no	Slry
--------	----------	------	----------	------

Write a SQL query to alter the employee table and add an attribute email to employee table.

ALTER TABLE table_name ADD column_name datatype;

ALTER TABLE Employee ADD Email varchar(5);

- **ALTER TABLE - ADD multiple Columns**
- To add multiple columns in a table, use the following syntax:
- **Syntax**

ALTER TABLE table_name ADD new_column_name column_definition

[FIRST | AFTER column_name], ADD new_column_name column_definition [FIRST | AFTER column_name],...;

SQL ALTER TABLE Statement

- **Write a SQL query to add height and weight fields to the existing table of employee.**

ALTER TABLE employee ADD height float AFTER Slry, ADD weight float AFTER Slry;

Emp_ID	Emp_Name	City	Phone_no	Slry	Email	weight	height
--------	----------	------	----------	------	-------	--------	--------

SQL ALTER TABLE Modify Column

- If you want to modify an existing column in SQL table, syntax is given below:

- **Syntax:**

ALTER TABLE *table_name* modify *column_name* *datatype*;

- **Write a sql query to change the email field length from 5 to 15.**

alter table employee modify email varchar(15);

SQL ALTER TABLE DROP Column

- Drop column using alter statement will delete a particular column.
- The syntax of alter table drop a column is given below:
- **Syntax:**

Alter table table_name drop column column_name;

Emp_ID	Emp_Name	City	Phone_no	Slry	Email	weight	height
--------	----------	------	----------	------	-------	--------	--------

Alter table employee drop column city;

Emp_ID	Emp_Name	Phone_no	Slry	Email	weight	height
--------	----------	----------	------	-------	--------	--------

SQL ALTER TABLE DROP Multiple Columns

- Drop column using alter statement will delete a particular column.
- The syntax of alter table drop a column is given below:
- **Syntax:**

Alter table *table_name* drop column *column_name1*, drop column *column_name2*,....drop column *column_nameN*;

Emp_ID	Emp_Name	Phone_no	Slry	Email	weight	height
--------	----------	----------	------	-------	--------	--------

Write an sql query to drop height and weight columns?

Alter table employee drop column height, drop column weight;

Emp_ID	Emp_Name	Phone_no	Slry	Email
--------	----------	----------	------	-------

SQL ALTER TABLE Rename column

- Using SQL Alter table rename, we can rename a column name.

- **Syntax:**

```
ALTER TABLE table_name CHANGE COLUMN old_name new_name  
column_definition [ FIRST | AFTER column_name ]
```

Write an sql query to change column name from Slry to Salary?

```
Alter table employee change column Slry Salary int;
```